Interactive Beat Annotation GUI

Table of Contents

Introduction

Contextualization and Problem Framing

Music Information Retrieval (MIR) is a research field that develops computational methods to analyze, interpret, and retrieve musical information from audio, examples include: harmony, melody, rhythm, and structural elements. Within MIR, Beat Tracking (BT) focuses on the automatic extraction of musical beat information from recordings. Since the beat provides a fundamental temporal framework for music, BT is a critical component of many MIR systems.

"Many MIR systems first analyze the beats as the starting point, assume the results would be correct, and use them as a unit for further processing." (Yamamoto, 2021)

The beat is typically defined as a regularly occurring pulse that underlies the temporal organization of a musical piece. However, the perception of the beat can vary across listeners. An objectively definitive beat can only be established if it is explicitly defined by the composer or agreed upon by performing musicians.

BT has been a research focus for decades. Existing algorithms perform reliably for music with steady, predictable tempos, such as Pop music. In contrast, music with expressive and/or fluctuating tempos remains challenging for autonomous beat detection. Examples of music genres that are still difficult to assess are, classical, jazz or world music. These can make the process of beat tracking more complex and ambiguous.

For this reason SoTA BT autonomous algorithms are all machine learning (ML) based. And although there are a few that are self-supervised (Desblancs et al., 2023), these are usually genre specific or are bound to certain musical configurations. For the cases that SoTA BT still can't solve, human-made beat annotation (or revisions) are still the most reliable.

Although there has been a lot of effort in advancing autonomous BT solutions, the same can not be said for human focused BT tools where it seems that they have been stayed largely unchanged for years now.

Research Questions

RQ1. What limitations exist in current beat annotation workflows when applied to challenging music with high expressiveness, such as tempo variability?

RQ2. How can different musical, model and signal representations support and guide the beat annotation process?

A RQ2 é intencionalmente abstrata, em termos práticos poderia ser partida em sub-questões:

- RQ2.1. How can a symbolic score representation, defined in musical time, be aligned with an audio-based representation defined in absolute time?
- RQ2.2. How can an audio spectrogram support accurate manual beat annotation in challenging music?
- RQ2.3. How can deep neural network output activations, representing the probability of beat presence over time, guide the annotation process?
- RQ2.4. Can such activation functions be used to define or support a beat quantization grid during annotation?

RQ3. To what extent does the proposed beat annotation workflow affect annotation time and efficiency when compared with existing tools?

State-of-the-Art

Beat tracking Research

It is clear that BT is one of MIR's most fundamental and most researched fields. As of now BT is considered a "solved" problem for music with steady 4/4 tempo, this includes a majority of western commercially available music (Country, Disco, HipHop, Rock, Metal, Pop etc.). Where SoTA BT algorithms still falter is in music that goes a stray fall from the never changing, 4/4 metric and steady bpm that is the standard for modern western music. In musical genres/examples with fluctuating bpm and complex or unconventional time signatures SoTA BT still falters. This includes music that is "acoustically" performed, meaning, music where the bpm can fluctuate for performative or expressive reasons. Examples include genres such as Classical, Jazz and World Music.

Besides this, the act of assessing the beat can be ambiguous. In some examples, annotators might tap at half, double, or even triple the actual beat. It is also possible for one to feel the subdivision at different levels, and this may impact the beat assessment of the annotator. For these cases, there can only be a "correct" assessment if the recording in question has an underlying musical score that explicitly describes the beat, i.e., classical music.

For these reasons, current SoTA beat tracking methods are predominantly based on neural networks (CNN, TCN's, Transformers). ZeroNS (Desblancs et al., 2023), BeatThis! (Foscarin et al., 2024), and HingeNet (Ru et al., 2025) are just 3 examples of SoTA algorithms that illustrate the complexity of BT:

- ZeroNS was the first self-supervised BT model. It works by analyzing separately the percussive (drums) and melodic parts of a given recording. A problem with this model is that by design is dependent of instrumentation, this model works best for music with drums and clear melodic instrument distinction (voice, guitar, keys, bass etc…). Besides this, ZeroNS performs poorly against other supervised SoTA models.
- BeatThis! Is still considered the SoTA BT model. It works by analyzing spectogram windows of the recording.
- HingeNet combines different BT models and ands a harmonically aware layer on top. Although HingeNet is newer, it is still F-measure testes show that it is slightly behind BeatThis!.

The performance of BT algorithms is based on an F-measure metric presented in 0.0 to 1.0 ratio or the equivalent in percentage. This measures a BT assessment against a ground-truth assessment of a given dataset. It takes into account correct beats, false positives (extra beats) and false negatives (missed beats), it also allows for a ±70ms window of tolerance for each beat.

BT models are tested against well documented datasets. In Figure 1 I present four that are common to see as benchmark tests for BT:

- GTZAN was specifically designed for musical genre classification. It is a collection of 2000 music excerpts of 30sec each. It covers 10 different music genres, plus 4 and 6 sub-genres of classical and jazz music respectively, each having 100 excerpts. This gives a total of 19h of audio to be tested.
- Ballroom is a dataset consisting a total of 698 total examples. With eight different dance music genres: Cha-Cha, Jive, Quickstep, Rumba, Samba, Tango, Viennese Waltz, and Slow Waltz. BPM ranges from 60-224. As this dance-music, bpm stays mostly the same across the excerpts, for this reason it is expected for SoTA models to perform F-measures > 90%.
- Hainsworth: A dataset consisting of 221 musical examples from a variety of genres, rock/pop, dance, classical, folk and jazz.
- SMC Mirex: Compiled to include pieces that are challenging for SoTA benchmark test. This data set consists of 217 manually-annotated highly precise musical pieces.

Figure 1: Tabel comparing the performance (F-measure in %) of SoTA BT models.

Dataset	Zero-Note Samba (2023)	Beat This! (2024)	HingeNet (MERT-based) (2025)
GTZAN	87.5%	89.1%	89.7%
Ballroom	91.1%	97.5%	97.3%
Hainsworth	78.3%	91.9%	91.4%
SMC Mirex	52.0%	62.7%	61.7%

It is important to note the distinction between Beat Tracking (BT) and Beat Annotation (BA).

- BT refers to autonomous beat prediction models.
- BA refers to a human-made annotation, as in using SV for doing a beat assessment. BA might use a BT algorithm as a starting point and then go from there.

Beat Annotation Methods

- Need for correction and shifting operations, ELABORAR MAIS

For music in which autonomous SoTA BT systems aren't efficient enough, relaying on human made beat annotations still might be preferable to the annotator:

"*A common method to create beat annotations for music recordings is to let a human annotator tap along with them.*" (Driedger et al., 2019)

The problem with this, is that these annotations will probably not be accurate enough even if the user preforming them is an experienced musician. There are multiple reasons that stem from perception, human motor noise, jitter and system latency (Driedger et al., 2019). Thus when tapping, there is the need to shift marked beats in order to make them accurate:

"*This shifting operation is particularly relevant when correcting tapped beats, which can be subject to human motor noise as well as jitter and latency during acquisition.*" (Sá Pinto et al., 2020)

Existing Tools for Beat Annotation

- Emphasize *design intent mismatch* (general-purpose vs BT-specific), mmm será mesmo necessário?

Sonic Visualizer is still the go-to software for scientific and MIR purposed audio analysis. However when it comes to BT, SV is outdated due to it's inefficient BT labeling process, that is still very much manual and cumbersome (Yamamoto, 2021). SV also is heavily reliant on VAMP plug-ins, modern SoTA BT algorithms are complex ML applications that may be difficult to implement into VAMP plug-ins and/or there is no direct or valued incentive for making these algorithms available in this way.

Time Instances inside SV (name of the main tool in SV for BT), can be exported and imported as simple .txt files. Thus, for one to benefit from a SoTA BT algorithm, one needs to follow a few steps:

- First, one needs to have access the algorithm in itself which might involve not only browsing/researching, but also probably, accessing some publicly available source-code repository.
- Secondly, one needs to know not only how to compile and run the program, but also export the retrieved data into a .txt file that SV can read.
- Besides this, we also need to take into account that to accomplish these steps, one needs at least, to be minimally literate/experienced with programming.

A problem with this, is that other users, who may not be experienced in programming or literate in MIR research, will rely on the most easily accessible BT algorithm inside SV, and these will probably be Beat Trackers like: BeatRoot (Dixon, 2001), Tempo and Beat Tracker - Queen Mary University of London (Stark et al., 2009), and IBT – INESC Beat Tracker (Oliveira et al., 2010) among others. The reason being that, although they don't come pre-installed with SV, they can be added via a native feature that allows to browse and install VAMP plugins without exiting the program (referring to the "Find Transform" SV tool). Crucially it is important to note that, as of the writing of this paper, you will not find any SoTA BT using this SV built in feature, in fact one will only find outdated BT that are at least more then a decade old.

Falta aqui algo para ligar estas partes…

One of the most common autonomous BT assessment mistakes are the Double/Half time errors, in which the algorithm might mark the beat in orders of integer multiplication (or division) from the actually correct beat.

Another common mistake might be inducing multiple closely sequenced markings in the place where only one should be, or the contrary, omitting/skipping beats. For correcting these type of mistakes, current Beat Tracking software only allow correcting beats one-by-one. A simple workaround for this mistake would be to develop a feature in which the user could select a time region of beats and perform an operation such as:

'In this region, that start at xx:xx and ends at yy:yy , the tempo is a steady X bpm with Y/Z timeSignature.' The program would then shift/eliminate/highlight marked beats that deviate from the "user statement".

HIL Approach for BT

In 2021 Yamamoto proposed a Human-In-the-Loop (HIL) approach for BT, the problem is, although this paper seems promising, the software developed for this paper is closed-sourced, property of Yamaha Corp. and thus, it is not likely to be publicly accessible.

The paper also fails to mention how a spectrogram can be beneficial as opposed to the more conventional waveform representation of audio. A waveform can be useful for identifying general differences in energy, on the other hand spectograms can, besides that, represent information related with spectral energy in time (frequency-energy).

A clear example of this would be: In a waveform, one can distinguish a *ff* section of from a *pp* in a recording. In a spectogram, one can not only distinguish these, but also understand if a particular section would be chord hit, or an *arpeggio,* along side noticing where these note onsets would be located pitch wise.

Audio-to-Score Alignment for BT

Something that is not very well documented, is the relevance of following along a music score when performing a beat annotation. Besides helping the annotator follow along more accurately the recording, a score might notice the annotator in advance of any bpm or time signature changes. Besides that, when using a Spectrogram-Based view of the recording, one can distinctively align a specific onset to a concrete and exact moment in the recording timeline, this is essential to know if a beat is accurately assessed or not.

Piano Precision (Jiang et al., 2025), is an example of a SV fork that was designed for displaying musical scores alongside audio recordings in a unified graphical interface. This program works surprisingly well for classical Solo Piano pieces. Now there are two question that arrive from this:

- One is that, by following along the music sheet, Beat Tracking autonomously should be as straight forward as making the program align a beat marking with the respective note onsets that are marked in the score. Although Piano Precision already does this with high enough accuracy for Audio-to-Score alignment, for beat tracking adjustments might be needed, yet, it still might give a better starting point than some SoTA beat assesments.
- The other would be to integrate this idea of Audio-to-Score alignment with a Spectral View of the recording. In Piano Precision this is done via a side-window that displays the a single A4 page of the score, one at a time. When the user hits play, a darkened highlight displays the current playback position in the score, while in another window a spectral view of the audio is following along the playback position horizontally. This allows for a musically natural follow-along of the score, mimicking how a musical score is actually read. However, for audio analysis and visualization purposes, it could be beneficial for the score to be aligned and viewed along side the playback position, meaning, horizontally and with note onsets that are aligned with their respective absolute position in the timeline. For BT this could allow the user to “grab a beat” and position it on the it’s corresponding position on top of the spectogram representation. Also this could be beneficial for more easily detecting beats that should be shifted.

Identified Gaps

Add a synthesis subsection (currently missing)

This is crucial.

Summarize:

- Tooling gap
- Interaction gap
- Accessibility gap
- Visualization gap

This subsection should directly justify your contribution

Work Plan

12/February:	Submission of state-of-the-art report
20 to 24/April:	Intermediate presentation
15/June:	Submission of thesis for supervisor validation (SIGARRA)
30/June:	Validation by supervisors (SIGARRA)
13 to 24/July:	Public presentation/Defence
31/July:	Final version submitted (SIGARRA)

Bibliography

Desblancs, D., Lostanlen, V., & Hennequin, R. (2023). Zero-Note Samba: Self-Supervised Beat Tracking. *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, *31*, 2922–2934.

Dixon, S. (2001). An interactive beat tracking and visualisation system. *Proceedings of the International Computer Music Conference (ICMC)*, 215–218.

Driedger, J., Schreiber, H., de Haas, W. B., & Müller, M. (2019). Towards Automatically Correcting Tapped Beat Annotations for Music Recordings. *Proceedings of the 20th International Society for Music Information Retrieval Conference (ISMIR)*.

Foscarin, F., Schlüter, J., & Widmer, G. (2024). Beat this! Accurate beat tracking without DBN postprocessing. *Proceedings of the 25th International Society for Music Information Retrieval Conference (ISMIR 2024)*.

Jiang, Y., Weigl, D. M., & Goebl, W. (2025). Piano Precision: Advancing Music Performance Analysis by Integrating User-Correctable Audio-to-Score Alignment. *Proceedings of the International Computer Music Conference (ICMC)*.

Oliveira, J. L., Gouyon, F., Martins, L. G., & Reis, L. P. (2010). IBT: A Real-Time Tempo and Beat Tracking System. *Proceedings of the 11th International Society for Music Information Retrieval Conference (ISMIR)*.

Ru, G., Wang, J., Zhao, J., Wu, Y., Yu, Y., Jiang, N., Wang, W., & Li, W. (2025). *HingeNet: A Harmonic-Aware Fine-Tuning Approach for Beat Tracking* (arXiv:2508.09788). arXiv. https://doi.org/10.48550/arXiv.2508.09788

Sá Pinto, A., Domingues, I., & Davies, M. E. P. (2020). Shift If You Can: Counting and Visualising Correction Operations for Beat Tracking Evaluation. *Extended Abstracts for the Late-Breaking Demo Session of the 21st International Society for Music Information Retrieval Conference (ISMIR)*.

Stark, A. M., Davies, M. E. P., & Plumbley, M. D. (2009). Real-time beat-synchronous analysis of musical audio. *Proceedings of the 12th International Conference on Digital Audio Effects (DAFx-09)*.

Yamamoto, K. (2021). Human-in-the-Loop Adaptation for Interactive Musical Beat Tracking. *Proceedings of the 22nd International Society for Music Information Retrieval Conference (ISMIR)*.